The Dependence of Bare Soil Albedo on Soil Water Content[1]

S. B. IDSO, R. D. JACKSON, R. J. REGINATO, B. A. KIMBALL AND F. S. NAKAYAMA

U. S. Water Conservation Laboratory, Phoenix, Ariz. 85040

(Manuscript received 12 April 1974, in revised form 30 July 1974)

ABSTRACT

Simple albedo measurement may prove useful for sensing surface soil water content and as a research tool in the study of evaporation of water from soil. Intensive concurrent measurements of the albedo and soil water content of a drying bare soil indicate that albedo, normalized for sun zenith angle effects, is a linear function of the soil water content of a very thin surface layer (less than 0.2 cm thick) over a sizeable volumetric water content range (0.00 to 0.18 for an Avondale loam). Albedo is also well correlated with the average soil water content of greater soil thicknesses. Measurements to a depth of 10 cm indicate that the relation is relatively independent of season.

1. Introduction

One of the more practical uses presently sought for various remote sensing techniques is that of assessing the soil water content of various land surfaces. Schmugge *et al.* (1974) discuss the present status of adapting microwave radiometers to reach this goal; while Doll (1973) discusses the possibility of using the polarization percentage of reflected light, and Skidmore *et al.* (1973) recount some success utilizing a reflectometer sensitive to selected near-infrared wavelengths. In this paper, we consider the possibilities for total albedo measurements to provide complementary data and extend the capacity of researchers to acquire these data via a relatively simple technology.

It has long been recognized that bare soil albedo, defined as the ratio of reflected to incoming solar radiation, is dependent upon the moisture status of the soil surface (Ångström, 1925; Büttner and Sutter, 1935; Brooks, 1952). Little thought was given to the predictive value of this relation, however, until Bowers and Hanks (1965) conducted a detailed laboratory study of the spectral reflectance properties of several soils at different water contents. Although their results indicated that there was a good potential for reflectance measurements to specify surface soil water content, the practicality of the technique was not thoroughly explored. Coulson and Reynolds (1971) obtained detailed spectral reflectance measurements of a wet and dry Yolo loam soil in the field, but included no actual measurements of the soil water status; and Tuller and van Heuklon (1972) made several sets of energy balance measurements from the ocean surf up the beach to dry sand, but did not explicitly specify the albedo or present detailed profiles of the soil water content. Thus, to better evaluate the potential for albedo measurements to estimate soil water content, we conducted a series of experiments involving intensive concurrent measurement of albedo and soil water content under natural field conditions during several seasons of the year. These experiments, together with their results, are described below.

2. Experiments

Four separate experiments were conducted on an Avondale loam soil [fine-loamy, mixed (calcareous), hyperthermic, Anthropic Torrifluvent] at Phoenix, Ariz., in May, July, September and December 1973. In each of these experiments, the smooth, bare field (72 m by 90 m) was initially irrigated with approximately 10 cm of water and then allowed to dry. Every 20 min during these drying periods, measurements were made of incoming solar radiation with an upright Eppley[2] pyranometer and reflected solar radiation with an inverted Eppley pyranometer and an inverted Kipp solarimeter in two widely spaced locations. Soil water contents were measured gravimetrically in areas similar to those viewed by the inverted pyranometers by sampling the 0–0.2, 0–0.5, 0–1, 1–2, 2–4, 4–6, 6–8 and 8–10 cm depth increments at the same 20-min intervals in six different locations. The albedo measurements were started before irrigation in each instance and extended for at least two weeks thereafter. Soil moisture sampling, however, was not begun until some drying had occurred, permitting access to the field. The soil was generally at a volumetric water content of 0.30 at

[1] Contribution from the Agricultural Research Service, U. S. Department of Agriculture.

[2] Trade names are included for the benefit of the reader and imply no endorsement or preferential treatment of the products listed by the U. S. Department of Agriculture.

Unauthenticated | Downloaded 06/01/24 09:13 PM UTC

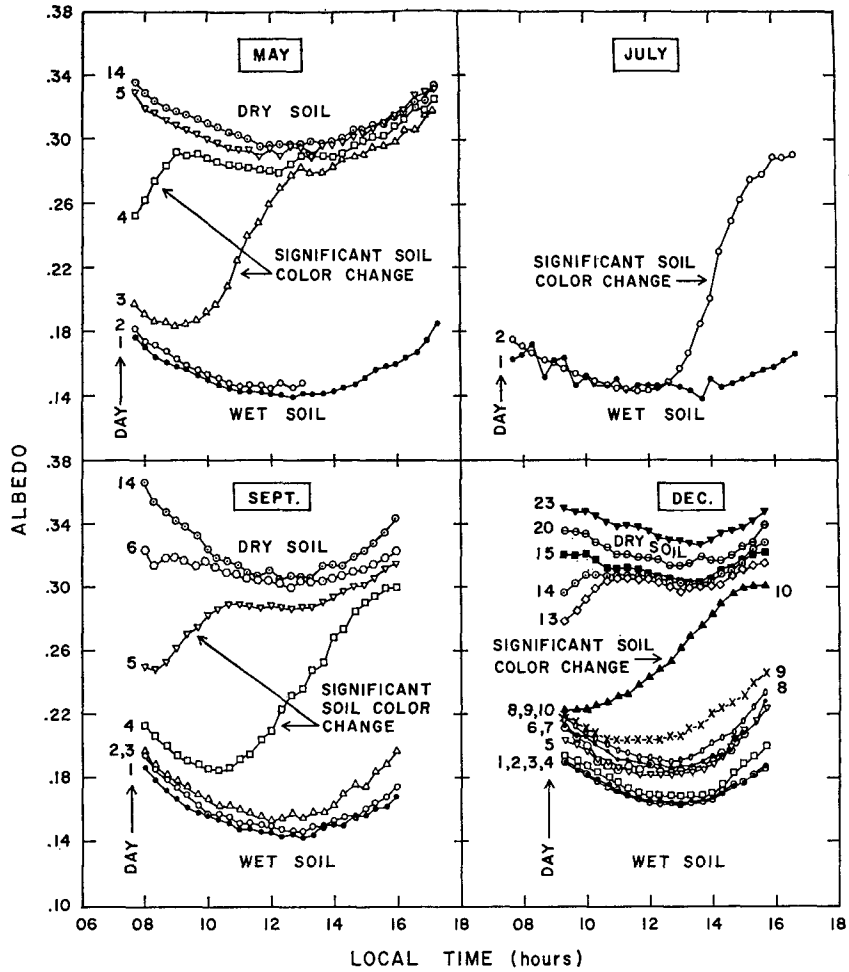

FIG. 1. Diurnal variations of albedo for specified days after irrigation of a smooth bare Avondale loam soil in May, July, September and December 1973 at Phoenix, Ariz. Munsell color notation supplied for this soil by the Soil Conservation Service is 10YR 5/3 when dry and 10YR 3/3 when wet.

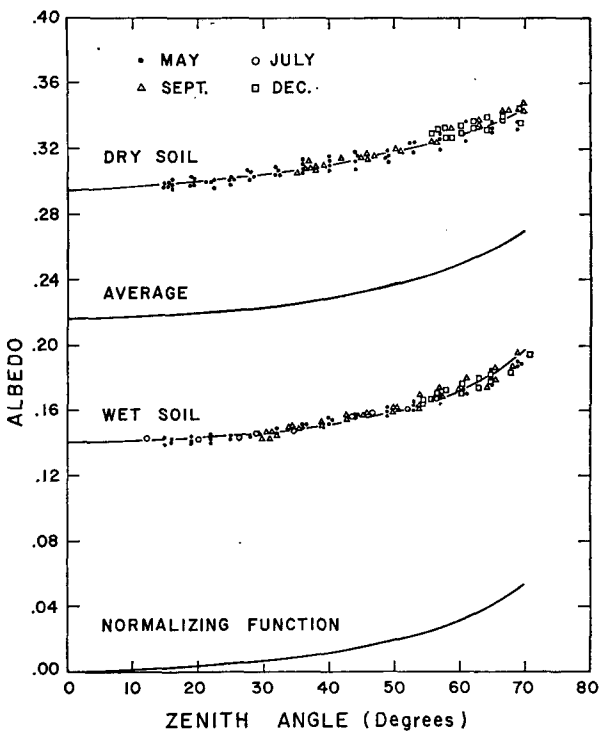

FIG. 2. Wet and dry soil albedo vs zenith angle for the four experiments, plus the normalization function derived therefrom to remove zenith angle effects from all of the raw albedo data.

this stage; and sampling was continued until the water content of the 0–0.2 cm layer was reduced to about 0.03. The procedures followed in all experiments were identical, except that the May study also included albedo measurements of an area of raked soil having roughness elements on the order of 1 cm height. More details on the soil sampling techniques and the water content calculations are given by Jackson *et al.* (1973). A paper describing the very thin section sampling is being prepared.

3. Results and analyses

Albedo measurements of the smooth soil at various times after irrigation are presented in Fig. 1 for all four experiments. The July data cover only two days, due to an untimely thunderstorm which terminated that experiment on the evening of the second day after irrigation.

The data in Fig. 1 show that there are three major regions of characteristic albedo variation with time through the day. The first is the initial stage when the surface soil is rather wet (volumetric water content >0.20), and the change in albedo with time appears to be symmetrical about solar noon in response to zenith angle effects. The second is the intermediate stage when rapid drying of the surface is occurring and the value of the soil color is visibly changing from dark to light,

Unauthenticated | Downloaded 06/01/24 09:13 PM UTC

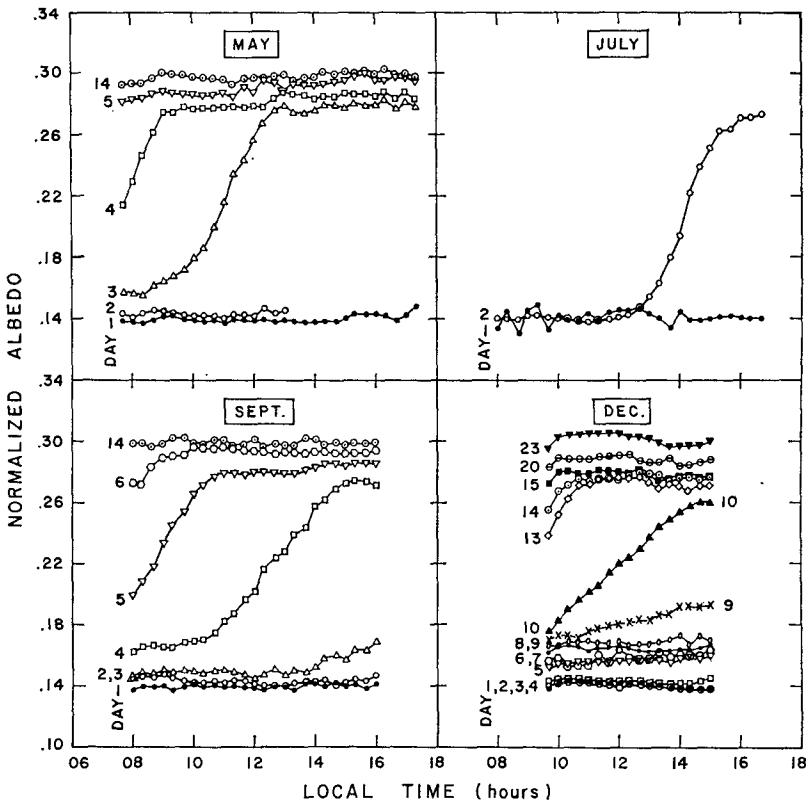

FIG. 3. Diurnal variations of normalized albedo for specified days after irrigation of a smooth bare Avondale loam soil in May, July, September and December 1973 at Phoenix, Ariz.

dramatically raising the albedo. Finally, the third stage occurs when the surface has become rather dry (volumetric water content <0.04), and the change in albedo with time is again symmetrical about solar noon.

To relate albedo solely to soil water content, it is necessary to remove zenith angle effects. Thus, the albedo data in Fig. 1 for both the wet and dry ranges defined above are replotted in Fig. 2 as a function of the sun's zenith angle. Two important results are there observed. First, the data from all four times of year describe the same two unique curves. Second, these two curves, for wet and dry conditions, are practically identical in shape. Thus, their average was taken and normalized to zero at zero zenith angle. Subsequently, the appropriate values of this normalizing function were subtracted from the raw albedo data of all of the experiments corresponding to the same zenith angles to yield the normalized albedo plots of Fig. 3. There it is observed that all of the experiments give consistent results in terms of albedo absolute magnitude. Each data set begins at an initial normalized albedo value of 0.14 and approaches an upper limit somewhere just above 0.30.

The albedo of the raked soil in the May experiment exhibited the same type of variation with time as did the smooth soil, although its absolute magnitude was somewhat lower. Comparative plots of the smooth and raked soil normalized albedos in Fig. 4 indicate that this difference varied from about 0.02 when the soil was very

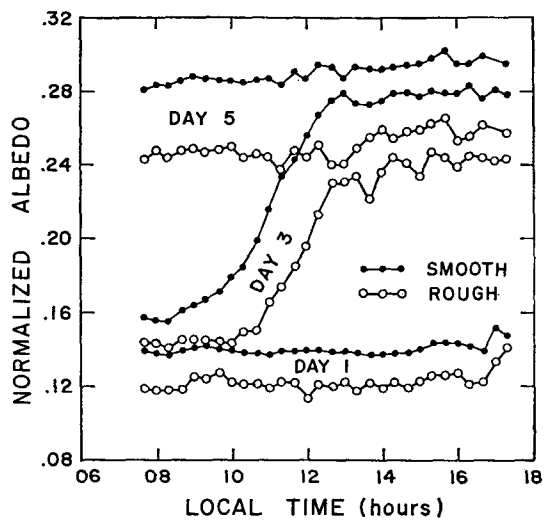

FIG. 4. Diurnal variations of normalized albedo for the 1st, 3rd and 5th days after irrigation of a smooth and rough bare Avondale loam soil at Phoenix, Ariz., in May 1973.

Unauthenticated | Downloaded 06/01/24 09:13 PM UTC

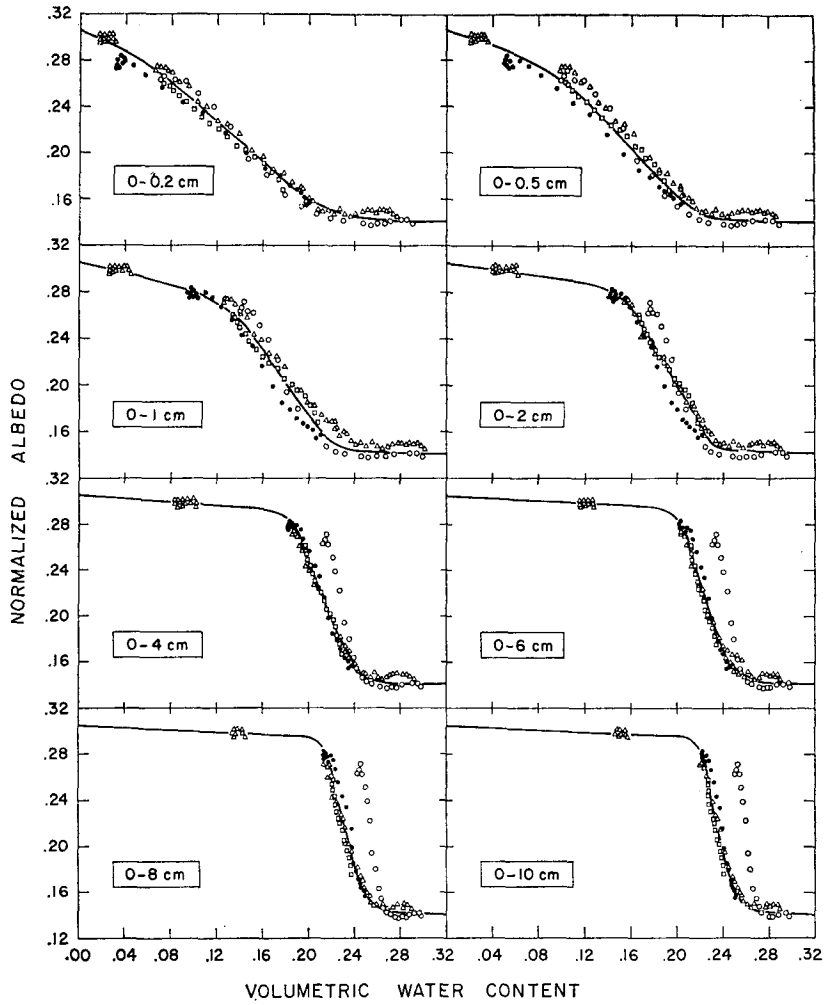

FIG. 5. Normalized albedo vs average volumetric water content of eight different soil layers having the surface as their upper boundary: solid circles, May; open circles, July; triangles, September; squares, December.

wet to 0.04 once it started to rapidly dry on the third day after irrigation. Self-shading effects appear to be responsible for these differences, as they would be expected to increase in magnitude as the soil became lighter with drying.

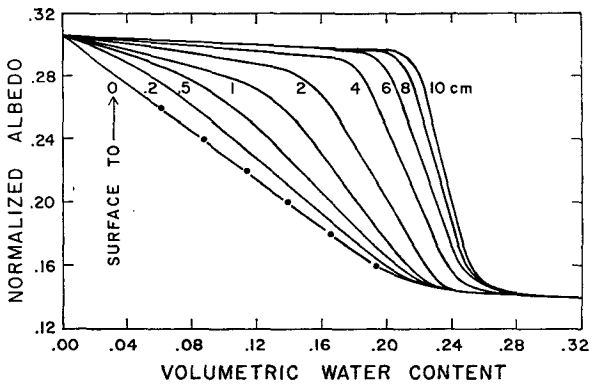

FIG. 6. Normalized albedo vs average volumetric water content of nine different soil layers having the surface as their upper boundary.

The normalized albedo values of Fig. 3 are correlated in Fig. 5 with several measures of the average soil water content from the surface down to various depths terminating at 10 cm. As can be seen in these plots, the data for the four different times of year all give consistent results—except for the July correlations with water content averages extending to depths >2 cm. Due to the extremely rapid drying conditions in July, forcing us to make soil water content measurements on the second day after irrigation, gravity drainage of water from the 2–10 cm region was not complete. Thus, this region of the soil had a proportionally greater water content on day 2 in July than on day 3 in May, day 4 in September, and day 10 in December, when soil water sampling was conducted in those experiments.

The several curves of Fig. 5 depicting the normalized albedo-soil water content relationships defined by all

Unauthenticated | Downloaded 06/01/24 09:13 PM UTC

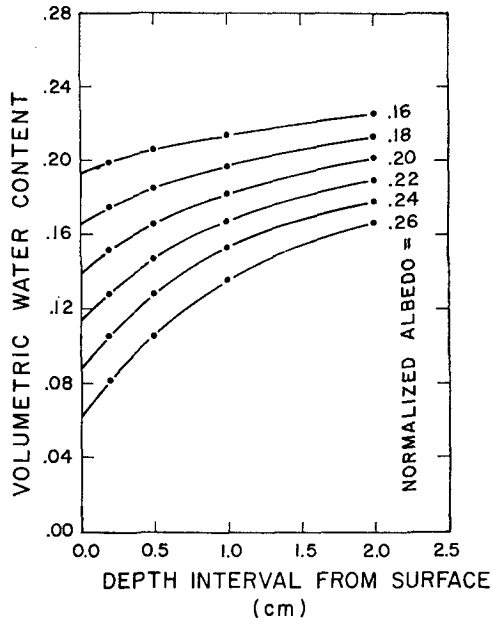

FIG. 7. Average volumetric water content vs depth of averaging for six selected constant values of normalized albedo.

four experiments for the upper 2 cm of soil and by the May, September and December experiments for the lower 8 cm of soil are combined in one set in Fig. 6. Values of the volumetric soil water content were extracted from these plots for several constant albedo values at the various depth interval lines and plotted versus this latter parameter in Fig. 7. Smooth lines were drawn connecting these points and extrapolated to yield the volumetric soil water contents at the very surface of the soil for the chosen constant normalized albedo values. These albedo-water content points were then replotted on Fig. 6 and the relationship between them for the very surface of the soil determined. As can be seen there, it was completely linear over a water content range from 0 to 0.18.

4. Concluding discussion

The results of these experiments indicate that simple albedo measurement may provide a reasonable means for the remote sensing of soil water content via readily available technology. Albedo, when normalized to remove sun zenith angle effects, appears to be a linear function of the water content of the uppermost layer of the soil. For soil depths on the order of 2 cm, the albedo-soil water content relationship appears to be independent of season. For greater depth intervals, a given albedo may correspond to a slightly greater average soil water content during periods of extremely high evaporative demand. These results should be general in nature for all soils. Differences in specifics such as initial and final absolute magnitudes, however, will need to be determined for each individual soil type and surface condition.

The finding that albedo measurements can be used to estimate the water content of the very surface of the soil also has important consequences for studies we and others are conducting on ways to more efficiently utilize water in agriculture. Since knowledge of the surface soil water content will enable calculations of soil water flux to be made very near the soil-air interface, a greater understanding of the coupling of evaporative water loss with the properties of the soil surface and the supply of water to it from below should ensue. Instrumentation for albedo measurement is widely available and should allow many investigators to use this approach in evaporation studies.

REFERENCES

Ångström, A., 1925: The albedo of various surfaces of ground. *Geograf. Ann.*, **7**, 323–342.

Bowers, S. A., and R. J. Hanks, 1965: Reflection of radiant energy from soils. *Soil Sci.*, **100**, 130–138.

Brooks, F. A., 1952: Atmospheric radiation and its reflection from the ground. *J. Meteor.*, **9**, 41–52.

Büttner, K., and E. Sutter, 1935: D. Abkühl. grösse in d. Dünen. Rückstrahl. verschied. Bodenbedeck. f. uv. u. gesamte Sonnenstrahl. *Strahlentherapie*, **54**, 156–173.

Coulson, K. L., and D. W. Reynolds, 1971: The spectral reflectance of natural surfaces. *J. Appl. Meteor.*, **10**, 1285–1295.

Doll, B., 1973: The potential use of polarized reflected light in the remote sensing of soil moisture. Rept. No. ECOM-5501, Army Electronics Command, Fort Monmouth, N. J., 1–19.

Jackson, R. D., B. A. Kimball, R. J. Reginato and F. S. Nakayama, 1973: Diurnal soil-water evaporation: Time-depth-flux patterns. *Soil Sci. Soc. Amer. Proc.*, **37**, 505–509.

Schmugge, T., P. Gloersen, T. Wilheit and F. Geiger, 1974: Remote sensing of soil moisture with microwave radiometers. *J. Geophys. Res.*, **79**, 317–323.

Skidmore, E. L., J. D. Dickerson and H. Schimmelpfennig, 1973: Evaluating surface soil water content by reflectance measurement. *Agron. Abstracts*, Annual Meeting Amer. Soc. Agron., 75.

Tuller, S. E., and T. van Heuklon, 1972: The relation of net and solar radiation over bare sand surfaces. *Arch. Meteor. Geophys. Bioklim.*, **B20**, 179–190.

Unauthenticated | Downloaded 06/01/24 09:13 PM UTC